Technological Rivalry and the Reorganization of Strategic Trade Networks

Md Mohiuddin*

Abstract

How do geopolitical technology shocks reorganize strategic trade networks? This paper examines whether rising exposure to artificial intelligence (AI) technologies reshapes the geopolitical structure of U.S. supply-chain dependence. I construct new measures of AI-related trade exposure by combining patent-based technology intensity with sectoral trade composition.[1] Using panel fixed-effects models, dynamic event-study specifications, local projection impulse responses, and Bayesian vector autoregressions, I document three findings. First, U.S. AI-related trade networks experienced substantial geopolitical reallocation following the escalation of U.S.–China technological rivalry after 2018. Second, baseline regime-interaction estimates indicate that periods of rising AI exposure were initially associated with greater China-linked import dependence relative to the pre-2016 period. Third, dynamic responses reveal persistent supply-chain adjustment over medium horizons: following AI exposure shocks, China-linked trade dependence gradually declines while diversification toward alternative supplier networks increases. Forecast-error variance decompositions further show that AI exposure shocks explain a substantial share of medium-run fluctuations in strategic trade concentration. Taken together, the findings suggest that technological rivalry reshapes global supply networks through dynamic reorganization rather than static trade contraction alone. The results highlight how strategic technology competition can alter the resilience and diversification structure of modern production networks.

JEL Codes: F14, F15, O33, F51, C33

Keywords: Artificial Intelligence; Technological Rivalry; Trade Networks; Supply Chains; Geoeconomic Fragmentation; Strategic Trade Reallocation

* Department of Economics, Texas Tech University, Lubbock, TX 79409. Email: mdmohiud@ttu.edu.

[1] Recent work shows that frontier technologies reorganize global production networks and alter strategic dependencies; see, e.g., Baldwin (2016), Autor et al. (2020), and Aghion et al. (2021).

Introduction

Technological rivalry is reshaping the organization of global trade networks in ways that traditional volume-based measures fail to capture.[2] Artificial intelligence (AI) technologies now constitute a strategic layer of modern production systems, embedding advanced semiconductors, precision machinery, and digital infrastructures into global value chains.[3] As geopolitical tensions intensify, trade in AI-related goods increasingly reflects not only comparative advantage but also national security concerns, export controls, and strategic decoupling.[4] The economic consequences of these shifts depend less on aggregate trade volumes and more on *how strategic trade networks are reorganized* across geopolitically sensitive partner blocs.

This paper studies how AI-induced trade exposure reshapes the geopolitical organization and concentration structure of U.S. strategic import networks and what this implies for institutional resilience. A growing literature shows that network concentration amplifies systemic vulnerability by transmitting localized disruptions through interdependent supply chains.[5] Production-network models demonstrate that upstream shocks can propagate nonlinearly when linkages are highly centralized (Acemoglu et al., 2012). Recent work on global value chains further shows that geopolitical fragmentation can reconfigure trade flows, generating efficiency–security tradeoffs that alter macroeconomic stability (Antràs & Chor, 2022; Baldwin, 2022).

This paper contributes by showing that technological exposure can reshape the partner structure of strategic trade networks, even when aggregate trade volumes remain relatively stable. While existing research has examined trade wars, tariffs, industrial policy, and geoeconomic fragmentation, much less is known about how *technology-driven trade exposure* interacts with geopolitical realignment to reorganize trade-network concentration.[6] AI technologies are uniquely suited to amplify these dynamics because they operate as general-purpose technologies with high upstream centrality and limited short-run substitutability (Brynjolfsson et al., 2021). Consequently, rising exposure to AI-linked supply

[2]Recent work emphasizes how technological competition reshapes global trade networks and strategic supply chains; see Baldwin (2016), Aghion et al. (2021), and Autor et al. (2020).

[3]Artificial intelligence technologies rely heavily on upstream semiconductor and digital infrastructure supply chains, creating strong technological interdependencies across countries (Brynjolfsson, Rock, and Syverson, 2021; Acemoglu et al., 2022).

[4]Recent geopolitical tensions have led to increased export controls and strategic decoupling policies targeting advanced technologies (Farrell and Newman, 2019; Aghion et al., 2021; Baldwin, Freeman, and Theodorakopoulos, 2023).

[5]Carvalho, V. (2014). "From Micro to Macro via Production Networks." *Journal of Economic Perspectives.*

[6]While recent work documents the rise of geoeconomic fragmentation and supply-chain realignment, empirical evidence on how technological exposure reshapes partner concentration remains limited (Aiyar et al., 2023; Alfaro and Chor, 2023; Freund et al., 2023).

chains may generate persistent strategic dependence and network vulnerability that standard bilateral trade metrics cannot detect.

Motivated by these concerns, I ask three integrated research questions:

1. Does rising exposure to AI-related technology trade generate structural reallocation across China-linked, allied, and offshore supply networks?

2. Are observed changes driven by discrete policy shifts (e.g., trade wars and export controls) or by persistent network reorganization induced by technological rivalry?

3. Do technological exposure shocks propagate system-wide through strategic trade networks, thereby reshaping concentration dynamics and institutional vulnerability?

To answer these questions, I construct new measures of AI-related trade exposure by combining international trade data with technological classification metrics. I then evaluate how exposure affects the distribution of import shares across three partner blocs—China-linked, allied, and offshore—using panel event-study models with bloc and year fixed effects. This design identifies whether rising technological exposure produces *differential network reorganization effects* in geopolitically sensitive trade systems after removing time-invariant partner characteristics and common global shocks.

Beyond static associations, I examine dynamic propagation mechanisms using Local Projection impulse responses, which trace how exposure shocks reshape trade concentration and supplier reallocation over multi-year horizons (Jordà, 2005). To assess broader system interactions, I estimate a compact Bayesian vector autoregression linking trade exposure, network concentration, and supply-chain centrality, allowing technological shocks to propagate endogenously across strategic trade networks (Sims, 1980; Bańbura et al., 2010). This integrated framework distinguishes persistent structural reorganization from temporary policy disturbances.

I document three main findings. First, U.S. AI-related trade networks undergo substantial geopolitical reorganization following the escalation of U.S.–China technological rivalry after 2018. While early exposure periods coincide with elevated China-linked dependence, later periods exhibit gradual diversification toward alternative supplier networks. Second, regime-interaction estimates show that AI exposure significantly reshapes partner concentration patterns relative to the pre-2016 baseline, suggesting persistent structural adjustment rather than one-time tariff effects. Third, Local Projection and Bayesian VAR evidence reveal medium-run propagation through strategic trade networks: exposure shocks generate persistent changes in sourcing structures and network concentration over multi-year horizons. Forecast-error variance decompositions further indicate that AI exposure shocks explain a substantial share of fluctuations in strategic trade concentration.

These results contribute to three strands of literature. First, I extend production-network models by demonstrating how geopolitical alignment shapes strategic concentration dynamics within technology supply chains (Carvalho, 2014; Acemoglu et al., 2012). Second, I complement research on geoeconomic fragmentation by identifying network reorganization—rather than aggregate trade contraction alone—as a central mechanism of strategic adjustment (Antràs & Chor, 2022; Baldwin, 2022). Third, I connect technological change to institutional resilience by showing how exposure to general-purpose technologies reshapes systemic risk structures and supply-chain dependence (Brynjolfsson et al., 2021).

More broadly, the findings suggest that institutional resilience depends not only on diversification of aggregate trade volumes but also on diversification of *network topology* and supplier dependence. Policies aimed solely at reducing bilateral trade deficits may therefore overlook deeper structural dependencies embedded within strategic technology ecosystems.

Literature Review

This chapter contributes to five closely related strands of literature: production networks and aggregate vulnerability, global value chains and geoeconomic fragmentation, trade shocks and domestic adjustment, technology and strategic competition, and institutional resilience under systemic risk.

First, my analysis relates to the literature on *production networks and aggregate fluctuations*. A central insight of this literature is that economic shocks propagate through input-output linkages, and that network concentration can magnify aggregate vulnerability.[7] Subsequent work shows that the topology of production networks shapes shock transmission, persistence, and nonlinear amplification, particularly when upstream suppliers are highly central (Baqaee and Farhi, 2019; Baqaee and Farhi, 2020). This chapter extends that logic to strategic trade networks by asking whether AI-related exposure reorganizes concentration and supplier dependence across geopolitically sensitive partner blocs.[8]

Second, this paper speaks to the literature on *global value chains and trade fragmentation*.[9] Global value chain research emphasizes that modern trade is increasingly organized

[7]See Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012), "The Network Origins of Aggregate Fluctuations," *Econometrica*; and Carvalho (2014), "From Micro to Macro via Production Networks," *Journal of Economic Perspectives*.

[8]See also Acemoglu, Akcigit, and Kerr (2016) on innovation networks and Gabaix (2011) on granular shocks in production networks.

[9]Recent empirical evidence documents increasing geopolitical fragmentation of global value chains (Aiyar et al., 2023; Alfaro and Chor, 2023; Freund et al., 2023).

through cross-border production stages rather than final-goods exchange alone.[10] More recent work argues that geopolitical tensions, industrial policy, and national-security concerns are fragmenting these networks (Aldasoro, Fender, and Hardy, 2023; Gopinath, 2024). Related contributions highlight friend-shoring, strategic relocation, and resilience-oriented sourcing as emerging features of the post-2018 trade environment.[11] My contribution is to show that one important margin of fragmentation is not simply lower trade volumes, but the dynamic reorganization of geopolitical concentration within strategic technology networks.

Third, I build on the literature studying *trade shocks and domestic economic adjustment*. Work on import competition demonstrates that externally generated trade shocks can have persistent effects on domestic labor markets, local production structures, and political economy outcomes.[12] That literature typically focuses on employment and regional adjustment. By contrast, I focus on how technology-linked trade exposure reallocates *partner shares* across blocs, thereby reshaping the network structure of strategic dependence.

Fourth, this chapter contributes to the literature on *technology, innovation, and strategic competition*.[13] This chapter also relates to work on strategic trade and innovation rivalry, where technological leadership affects both market structure and geopolitical leverage.[14] AI functions as a general-purpose technology with broad implications for productivity, industrial structure, and geopolitical rivalry.[15] Recent research increasingly links technology leadership to industrial policy and national security (Grossman and Helpman, 2021; Farrell and Newman, 2019; Aharoni et al., 2023). In particular, Farrell and Newman (2019)

[10] See Baldwin (2016), *The Great Convergence*; Johnson and Noguera (2012), "Accounting for Intermediates," *Journal of International Economics*; Koopman, Wang, and Wei (2014), "Tracing Value-Added and Double Counting in Gross Exports," *American Economic Review Papers and Proceedings*; and Antràs and Chor (2022), "Global Value Chains," in the *Handbook of International Economics*.

[11] See Javorcik (2020), "Global Supply Chains Will Not Be the Same in the Post-COVID-19 World," in Baldwin and Evenett (eds.); Bonadio, Huo, Levchenko, and Pandalai-Nayar (2021), "Global Supply Chains in the Pandemic," *Journal of International Economics*; and Di Nino et al. (2022), ECB work on friend-shoring and supply-chain reallocation.

[12] See Autor, Dorn, and Hanson (2013), "The China Syndrome," *American Economic Review*; Acemoglu, Autor, Dorn, Hanson, and Price (2016), "Import Competition and the Great US Employment Sag of the 2000s," *Journal of Labor Economics*; Caliendo, Dvorkin, and Parro (2019), "Trade and Labor Market Dynamics," *Econometrica*; and Pierce and Schott (2016), "The Surprisingly Swift Decline of US Manufacturing Employment," *American Economic Review*.

[13] Artificial intelligence has been described as a general-purpose technology with broad macroeconomic implications for productivity and global competition (Brynjolfsson, Rock, and Syverson, 2021; Agrawal, Gans, and Goldfarb, 2019).

[14] See Grossman and Helpman (2021) for recent work linking innovation, industrial policy, and strategic competition; see also Farrell and Newman (2019) on how network centrality can create coercive geopolitical leverage.

[15] See Bresnahan and Trajtenberg (1995), "General Purpose Technologies," *Journal of Econometrics*; Brynjolfsson, Rock, and Syverson (2021), "The Productivity J-Curve," *American Economic Journal: Macroeconomics*; and Aghion, Antonin, and Bunel (2021), *The Power of Creative Destruction*.

argue that network centrality can become a source of coercive power in international political economy.[16] My results fit that broader framework by showing that AI-linked trade networks can generate shifting patterns of structural exposure and strategic dependence even when aggregate trade remains high.

Fifth, the paper contributes to the literature on *resilience, robustness, and supply-chain vulnerability*. A growing body of work distinguishes resilience—the capacity to absorb shocks and reconfigure—from robustness, or the ability to resist change.[17] This distinction matters because policies that reduce aggregate trade volumes may not reduce vulnerability if strategic supplier dependence remains concentrated or insufficiently diversified.[18] Related work on disaster shocks, pandemic disruptions, and industrial bottlenecks shows that resilience depends on redundancy, substitutability, and the decentralization of critical inputs (Bonadio et al., 2021; Evenett, 2022; Bems and Johnson, 2023).[19] My chapter complements this literature by quantifying how AI-driven trade exposure changes concentration risk and strategic supplier dependence in a geopolitical setting.

Methodologically, this chapter also relates to the literature on *dynamic causal effects and system propagation*. I use Local Projections to trace medium-run responses because they allow flexible dynamic estimation without imposing a highly restrictive VAR structure.[20] I complement this with a compact Bayesian VAR to capture feedback between exposure, network concentration, and strategic trade reallocation (Bańbura, Giannone, and Reichlin, 2010; Koop, 2013).[21]

Taken together, these literatures suggest that the key policy question is not whether trade declines under geopolitical rivalry, but how technologically strategic trade networks reorganize under geopolitical competition. Existing work shows that production networks transmit shocks, that geoeconomic fragmentation is increasing, and that AI is becoming central to strategic competition. What remains less understood is how AI-related trade exposure reshapes the distribution of partner dependence and strategic supplier concentration across geopolitical blocs, how persistent those changes are, and whether they alter the resilience and vulnerability structure of strategic trade networks. Rather than treating

[16]See Farrell and Newman (2019), "Weaponized Interdependence," *International Security*.

[17]See Miroudot (2020), Resilience versus Robustness in Global Value Chains, VoxEU; and Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021), Supply Chain Disruptions, *Quarterly Journal of Economics*.

[18]Recent work highlights the importance of redundancy and diversification in supply networks for economic resilience (Bonadio et al., 2021; Barrot and Sauvagnat, 2016).

[19]See Evenett (2022), work on fragmentation and industrial policy; and Bems and Johnson (2023), recent work on supply-chain disruptions and trade-network reconfiguration.

[20]See Jordà (2005), "Estimation and Inference of Impulse Responses by Local Projections," *American Economic Review*; Auerbach and Gorodnichenko (2012), *American Economic Journal: Economic Policy*; and Ramey (2016), work on macroeconomic shocks and LP methods.

[21]See Bańbura, Giannone, and Reichlin (2010), "Large Bayesian Vector Autoregressions," *Journal of Applied Econometrics*; and Koop (2013), *Bayesian Econometrics*.

strategic dependence as static, this chapter emphasizes how technological rivalry dynamically reorganizes supplier concentration across geopolitical blocs over time. This chapter provides evidence on each of those margins.

Road Map. Section I describes the data sources, variable construction, and measurement of AI-related trade exposure across geopolitical blocs using OECD TiVA input–output linkages, UN Comtrade trade flows, and AI patent indicators. This section also introduces the key outcome variables capturing China-linked trade dependence, geopolitical supplier concentration, and strategic trade-network organization. Section II presents preliminary descriptive evidence documenting the evolution of AI-related trade exposure and geopolitical trade-network reorganization patterns. In particular, the section highlights changes in China-linked trade shares, diversification dynamics, and strategic supplier reallocation following the technological and geopolitical shifts beginning in the mid-2010s, with a notable acceleration after 2018.

Section III develops the unified econometric framework. The analysis begins with baseline panel fixed-effects regressions identifying the relationship between AI-related technological exposure and geopolitical trade dependence across partner blocs. The framework then introduces regime-specific interaction terms capturing the geopolitical transition beginning around 2018. Estimation follows modern panel-inference practices using clustered standard errors and alternative bootstrap procedures to address small-cluster settings.[22]

Section IV extends the empirical framework to dynamic analysis. Local Projection methods are used to estimate horizon-specific responses of trade-network concentration and China-linked supplier dependence following AI exposure shocks.[23] Complementary system dynamics are examined using a compact Bayesian VAR framework that captures interactions between China trade share, AI exposure, and strategic trade concentration.[24] Forecast-error variance decompositions are used to quantify the relative importance of exposure shocks in driving medium-run trade-network reorganization dynamics.

Section V reports the main empirical results and a comprehensive set of robustness checks. These include alternative exposure measures, inference procedures, concentration controls, export-placebo tests, and break-year validation exercises. Additional robustness exercises examine whether the observed dynamics reflect persistent geopolitical trade reallocation rather than temporary policy disturbances. Heterogeneity tests further evaluate whether adjustment patterns differ across economies with high versus low AI exposure intensity.

Section VI discusses the economic interpretation of the findings and their implications for

[22] Cluster-robust inference follows the recommendations of [6] and the applied guidance of [12]. Bootstrap inference for clustered settings follows [11].

[23] Local projections follow the methodology introduced by [18].

[24] Bayesian VAR methods follow the large-scale macroeconomic framework discussed in [9].

global technology competition, supply-chain resilience, and strategic trade policy. The section links the empirical results to recent literature on geopolitical competition in global production networks, technological rivalry, and supply-chain reorganization under fragmentation pressures. Section VII concludes.

I. Data and Variable Construction

i) Data Sources

This chapter integrates three major data systems to measure technological trade exposure and strategic concentration. First, I use bilateral trade flows from the UN Comtrade database, which provides harmonized country–partner trade values across detailed product classifications.[25] I extract U.S. imports and exports for AI-related products at the HS classification level and aggregate them annually for the period 1995–2024. Second, I use the OECD Trade in Value Added (TiVA) database to account for global value chain linkages and to contextualize gross trade flows within value-added networks.[26] TiVA indicators allow me to interpret trade concentration as dependence on foreign production stages rather than only final-goods exchange. Third, I construct technology intensity measures using USPTO AI-related patent counts.[27] These data allow me to classify trade flows into AI-intensive categories that are most relevant for strategic technological competition.

Together, these sources allow me to map trade exposure, technological intensity, and network concentration within a unified framework.

ii) Partner Bloc Classification

To analyze geopolitical concentration, I group trade partners into three blocs:

- **China Bloc**: China and closely linked production networks,
- **Allied Bloc**: OECD-aligned industrial partners,
- **Offshore Bloc**: emerging and non-aligned production centers.

Allied partners include OECD economies and major strategic technology partners of the United States, while offshore economies represent emerging manufacturing hubs not

[25] UN Comtrade is widely used to construct bilateral trade exposure measures in international trade research (Feenstra, Romalis, and Schott, 2002; Johnson and Noguera, 2012).

[26] Value-added trade measures allow researchers to account for upstream and downstream production linkages embedded in global value chains (Koopman, Wang, and Wei, 2014; Johnson, 2018).

[27] Patent-based indicators are widely used to measure technological exposure and innovation intensity (Bloom, Schankerman, and Van Reenen, 2013; Aghion et al., 2021).

closely aligned with either geopolitical bloc. This classification follows recent geo-economic frameworks that study strategic decoupling, and bloc-based trade realignment. [28]

iii) Outcome Variable: Strategic Import Concentration

Let M_{bt} denote U.S. imports from bloc b in year t and M_t denote total AI-related imports. I construct bloc shares:

$$s_{bt} = \frac{M_{bt}}{M_t} \quad where, \quad b \in \{\text{China, Allies, Offshore}\} \tag{1}$$

This share captures how U.S. strategic dependence is distributed across geopolitical partner groups. [29]

iv) Key Explanatory Variable: AI Trade Exposure

I define AI trade exposure as the intensity of AI-related imports relative to the size of the domestic economy:

$$AIExposure_t = \log\left(\frac{\text{AI Imports}_t}{GDP_t}\right) \tag{2}$$

This measure captures the economy-wide penetration of AI-linked trade and follows standard exposure constructions. [30]

v) Network Concentration: Herfindahl Index

To capture systemic vulnerability arising from partner concentration, I construct a trade-network Herfindahl index:

$$HHI_t = \sum_b s_{bt}^2 \tag{3}$$

Higher values indicate greater concentration and reduced diversification. [31]

vi) Regime Indicators

To capture major geopolitical shifts affecting technology trade, I construct regime indicators corresponding to key policy and shock periods: (i) a pre-escalation environment before the U.S.–China trade conflict, (ii) the onset of strategic technology rivalry starting

[28] Recent research documents the emergence of bloc-based trade fragmentation and friend-shoring patterns in global supply chains (Farrell and Newman, 2019; Javorcik, 2020; Aiyar et al., 2023).

[29] Using trade shares rather than levels isolates structural reallocation across partners and removes scale effects driven by aggregate demand fluctuations (Carvalho, 2014; Acemoglu et al., 2012).

[30] This construction follows the exposure approach used in import competition and technology diffusion studies (Autor, Dorn, and Hanson, 2013; Acemoglu et al., 2016).

[31] Concentration measures such as the Herfindahl-Hirschman Index are widely used to capture systemic fragility in production and financial networks (Gabaix, 2011; Acemoglu et al., 2012).

in 2018, and (iii) the pandemic-related supply-chain disruptions starting in 2020. [32]

F. Descriptive Statistics

Table 1 China-linked import shares display growing dispersion, and AI exposure rises steadily with technology diffusion. The HHI shows increasing partner concentration, implying reduced diversification. The short time series motivates inference robust to small samples and clustered shocks.

Table 1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	25%	Median	75%	Max
ChinaShare	0.320	0.132	0.077	0.220	0.368	0.432	0.463
AlliesShare	0.441	0.159	0.294	0.317	0.357	0.545	0.779
OffshoreShare	0.239	0.053	0.144	0.209	0.224	0.247	0.385
HHI	0.398	0.083	0.334	0.354	0.358	0.402	0.634
AIExposure	-3.754	0.277	-4.213	-4.046	-3.685	-3.519	-3.361
AIExposure_alt	0.034	0.079	0.023	0.026	0.034	0.040	0.049
Unemployment	5.568	1.718	3.625	4.394	5.172	5.944	9.608
LFPR	64.694	1.925	61.675	62.858	65.054	66.479	67.108
ln(GDP)	30.453	0.195	30.066	30.311	30.451	30.601	30.782

Notes: AIExposure and AIExposure_alt are log ratios of AI imports to GDP. The table reports summary statistics for annual data from 1995–2024. ChinaShare, AlliesShare, and OffshoreShare measure partner-bloc import shares in AI-related trade; HHI captures trade concentration. AIExposure reflects intensity of AI-related trade. Controls include unemployment, LFPR, and log real GDP

Appendix Table A1 shows negative China–ally correlations and positive links between AI exposure, concentration, and macro activity, motivating macro controls.

[32] Recent work shows that trade wars and pandemic disruptions produced structural shifts in global supply chains and production networks (Bonadio et al., 2021; Evenett, 2022).

G. Descriptive Patterns and Stylized Facts

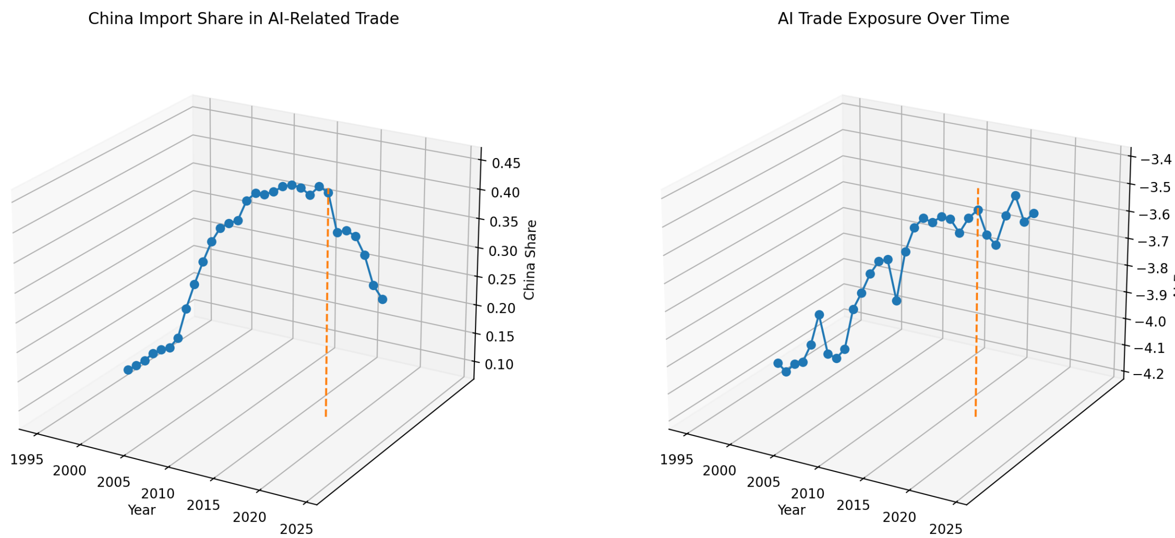

Figure 1: Strategic Import Concentration and AI Trade Exposure

Figure 1 visualizes the core empirical patterns motivating the analysis. China-linked import shares rise markedly after 2018, while AI trade exposure shows persistent growth across regimes. Concentration measures also trend upward, indicating increasing reliance on a narrower set of strategic partners. [33]

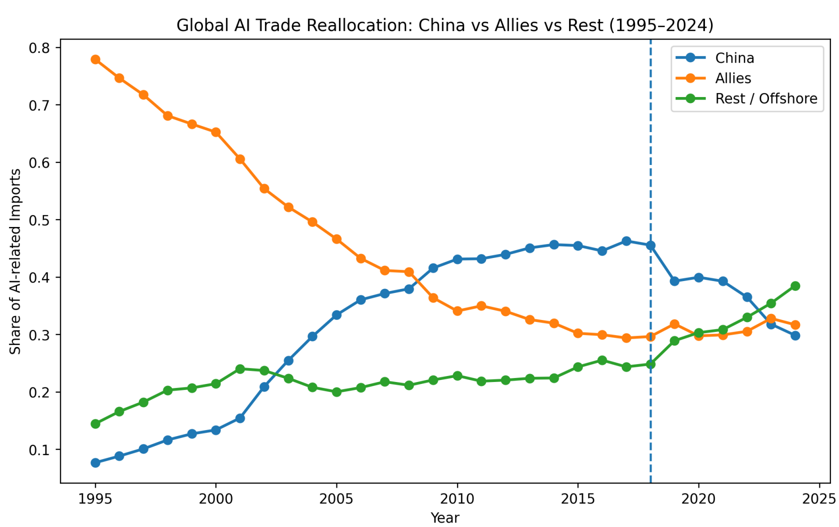

Figure 2: Global AI trade reallocation across geopolitical blocs. The figure reports the U.S. import share of AI-related trade sourced from China, allied economies, and offshore partners from 1995 to 2024. The dashed vertical line marks 2018, the onset of intensified U.S.-China technology rivalry.

[33] Recent studies document growing geopolitical realignment of trade flows and supply chains after the mid-2010s (Aiyar et al., 2023; Baldwin, Freeman, and Theodorakopoulos, 2023).

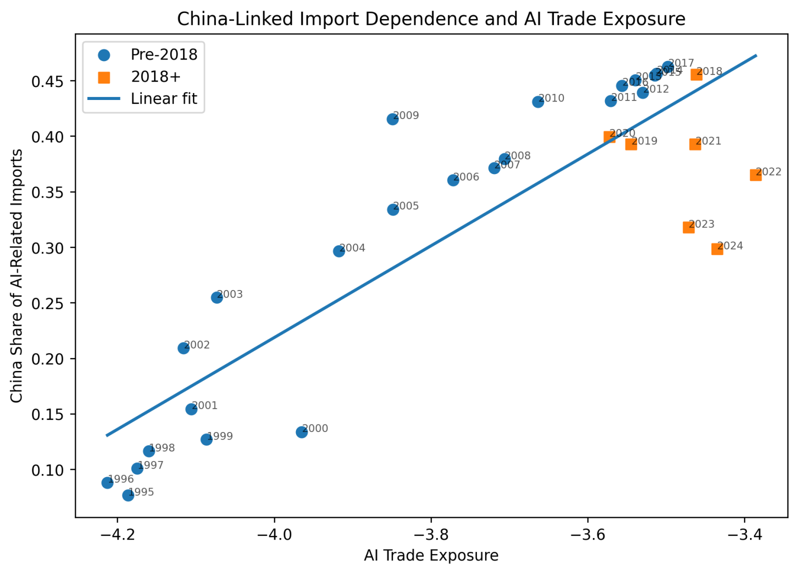

Figure 3: China-linked import dependence and AI trade exposure. Each point represents an annual observation. The figure plots the relationship between AI trade exposure and the China share of U.S. AI-related imports, with markers distinguishing the pre-2018 and post-2018 regimes. The upward association motivates the econometric analysis of whether rising technological exposure is linked to greater geopolitical concentration.

Figure 3 provides a reduced-form visual summary of the positive relationship between AI trade exposure and China-linked import dependence, particularly after the onset of intensified U.S.–China technological rivalry.

The descriptive patterns suggest a shift toward greater geopolitical concentration in AI-related trade networks. The next section formally tests whether these patterns reflect systematic responses to technological exposure shocks.

II. Econometric Framework and Identification

i) Baseline Identification Strategy

To identify whether rising technological trade exposure is associated with geopolitical concentration, I estimate a panel fixed-effects model that exploits variation across partner blocs and over time.

Let s_{bt} denote the share of AI-related imports sourced from partner bloc $b \in \{\text{China, Allies, Offshore}\}$ in year t. My baseline specification is:

$$s_{bt} = \alpha_b + \gamma_t + \beta_1 \left(AIExposure_t \times China_b \right) + u_{bt} \tag{4}$$

where α_b are bloc fixed effects capturing time-invariant structural differences across partner groups, and γ_t are year fixed effects absorbing common macroeconomic shocks. The interaction term isolates whether rising AI trade exposure is associated with a *differential reallocation* toward China-linked supply networks relative to allied and offshore partners. This specification can be interpreted as a difference-in-differences design in partner shares, where China-linked supply networks serve as the treated group and allied and offshore partners provide comparison groups (Carvalho, 2014; Acemoglu et al., 2012).

The empirical framework exploits variation in AI-related technological trade exposure across time and geopolitical blocs. The identification assumption is that changes in aggregate AI exposure primarily reflect global technological diffusion and innovation dynamics rather than endogenous bilateral supply-chain reallocation decisions. While reverse causality cannot be fully ruled out, several features mitigate this concern. First, the exposure measure aggregates imports across a broad set of AI-related products, capturing technology-wide diffusion rather than partner-specific sourcing. Second, regime interactions exploit the exogenous timing of geopolitical shocks, particularly the escalation of U.S.–China technology tensions after 2018. Third, placebo export regressions show no comparable patterns, suggesting the mechanism operates through import-side supply-chain adjustment rather than general trade expansion.

ii) Dynamic Event-Study Specification

Identification requires that, absent technological exposure shocks, the relative import shares of China-linked and non-China partner blocs would have followed parallel trends.
Identification requires that, absent technological exposure shocks, the relative import shares of China-linked and non-China partner blocs would have followed parallel trends.
To examine whether reallocation patterns coincide with geopolitical regime shifts, I estimate a dynamic event-study specification allowing the exposure effect to vary across geopolitical regimes:

$$s_{bt} = \alpha_b + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \left(AIExposure_t \times China_b \times D_\tau \right) + u_{bt} \tag{5}$$

where D_τ are regime indicators corresponding to:

$$R_t^{2016} = 1[t \geq 2016 \wedge t < 2018], \quad R_t^{2018} = 1[t \geq 2018 \wedge t < 2020], \quad R_t^{2020} = 1[t \geq 2020].$$

The year 2018 marks the onset of the U.S.–China technology conflict, including tariffs, export controls, and restrictions on Chinese technology firms. These policy shocks represent a structural break in global technology trade networks (Evenett, 2022). This staged treatment structure is consistent with modern event-style and dynamic-treatment frameworks.[34]

As a falsification exercise, I estimate the same specification using export shares. If the mechanism operates through strategic sourcing dependence, I should observe regime-specific China effects in imports but not in exports. This placebo design helps distinguish strategic import concentration from generalized trade expansion.

Finally, I evaluate whether reallocation toward China increases systemic exposure by estimating concentration equations using the Herfindahl-Hirschman Index. This links bloc-specific trade reallocation to a broader measure of strategic vulnerability.[35]

For inference, I report heteroskedasticity- and autocorrelation-consistent standard errors for aggregate time-series regressions and cluster-robust standard errors for bloc-level panel models.[36] Because cluster counts can be limited in some specifications, I also report restricted wild cluster bootstrap p-values for key interaction terms.[37]

This framework serves three purposes:

- Tests for pre-trends in partner reallocation,
- Identifies whether concentration accelerates after geopolitical escalation,
- Validates timing consistency of structural break patterns.

Event-study methods are widely used to trace dynamic adjustment paths around policy and geopolitical shocks (Autor et al., 2013; Jordà, 2005).[38]

iii) Local Projection Dynamics

Because technological trade exposure may propagate gradually through supply-chain networks, I estimate impulse-response functions using Local Projections:

$$s_{b,t+h} = \alpha_b + \gamma_t + \theta_h \left(AIExposure_t \times China_b \right) + X_t' \delta + u_{b,t+h} \tag{6}$$

for horizons $h = 0, \ldots, H$.

[34] For dynamic treatment-response logic, see Angrist and Pischke (2009) and Jordà (2005).

[35] HHI is a standard concentration metric in industrial organization and antitrust analysis; see DOJ and FTC (2010).

[36] For HAC inference, see Newey and West (1987). For cluster-robust inference, see Arellano (1987) and Cameron and Miller (2015).

[37] See Cameron, Gelbach, and Miller (2008).

[38] Event-study frameworks in panel settings allow visualization of dynamic treatment effects and pre-trend diagnostics (Sun and Abraham, 2021; Callaway and Sant'Anna, 2021).

Local projections estimate horizon-specific responses without imposing the dynamic restrictions inherent in vector autoregressions.[39] This framework identifies whether AI exposure shocks generate persistent trade-network reorganization, strategic supplier reallocation, and medium-run changes in China-linked dependence over time. Control variables include macroeconomic indicators such as unemployment, labor-force participation, and log real GDP.

iv) System Dynamics and Network Propagation

To assess broader propagation mechanisms, I estimate a Bayesian Vector Autoregression (VAR) system:

$$Y_t = A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + \varepsilon_t \tag{7}$$

where $Y_t = (ChinaShare_t, AIExposure_t, HHI_t)'$.

Impulse responses and forecast error variance decompositions reveal whether technological exposure shocks propagate through strategic trade-network reorganization and concentration dynamics rather than isolated bilateral adjustments.[40] The ordering assumes that AI exposure shocks affect trade concentration contemporaneously, while adjustments in concentration and supplier allocation occur gradually over subsequent periods. Residual diagnostics reported in Appendix Table A4 indicate no statistically significant serial correlation in the VAR residuals, suggesting that the lag structure adequately captures the dynamic adjustment process.

v) Endogeneity and Identification Assumptions

Identification relies on four assumptions.

First, bloc fixed effects remove time-invariant geopolitical alignment. Second, year fixed effects absorb global demand and macroeconomic shocks. Third, lagged exposure measures mitigate simultaneity between trade concentration and technological imports. Fourth, export-placebo specifications test whether estimated effects reflect general trade expansion rather than strategic concentration.

Together, these strategies parallel identification approaches in trade-shock and network-adjustment literatures (Autor et al., 2013; Acemoglu et al., 2016). A potential concern

[39] Local Projection methods estimate impulse responses without requiring full system specification (Jordà, 2005; Montiel Olea and Plagborg-Møller, 2021).

[40] VAR frameworks are standard for tracing dynamic propagation in macroeconomic systems (Sims, 1980; Diebold and Yilmaz, 2014). Bayesian shrinkage improves small-sample stability (Bańbura et al., 2010).

is reverse causality if rising concentration induces greater imports of AI-intensive goods. Lagged exposure measures help mitigate this concern by ensuring that trade exposure precedes observed reallocation.

vi) Heterogeneity and Robustness

I test heterogeneous effects across high- vs low-exposure periods, alternative partner classifications, concentration-adjusted specifications using HHI, and linear vs regime-based time trends. Robustness checks use heteroskedasticity- and autocorrelation-consistent errors, wild-cluster bootstrap inference, and alternative exposure constructions.[41] These tests ensure that findings are not driven by specific classification schemes or functional forms.

I allow treatment effects to vary with baseline exposure intensity:

$$s_{bt} = \alpha_b + \gamma_t + \beta_1 (AIExposure_t \times China_b) + \beta_2 (AIExposure_t \times China_b \times HighExposure_t) + u_{bt} \tag{8}$$

This specification evaluates whether China-specific dependence intensifies in already highly exposed environments, providing mechanism-consistent heterogeneity evidence. One interpretation is that economies already highly integrated in AI-related trade networks have partially diversified their supply chains prior to the geopolitical shock. In contrast, economies with lower initial exposure adjust more strongly once technological competition intensifies, generating larger observed reallocation.

To ensure results are not driven by functional form, I estimate alternative specifications:

- **Trend controls:** Linear and quadratic time trends.
- **Alternative exposure measure:** GDP-based AI penetration.
- **First-difference specification:**

$$\Delta ChinaShare_t = \kappa \, \Delta AIExposure_t + \eta_t$$

- **Event-study dynamics:**

$$s_{bt} = \alpha_b + \gamma_t + \sum_{k \neq -1} \beta_k (AIExposure_t \times China_b \times D_{t-2018=k}) + u_{bt}$$

[41] Cluster-robust inference and bootstrap procedures improve reliability in panels with few clusters (Cameron and Miller, 2015).

Event-study designs allow visualization of pre-trends and dynamic treatment paths.[42]

a) Export-Side Placebo Design

To test whether the results reflect strategic import dependence rather than general trade expansion, I estimate the same specification using export shares as the dependent variable:

$$x_{bt} = \alpha_b + \gamma_t + \theta_1(AIExposure_t \times China_b) + \sum_r \theta_r(AIExposure_t \times China_b \times R_t^r) + e_{bt} \quad (9)$$

If the mechanism operates through import-side strategic concentration, regime-specific China effects should be weaker or absent in exports. This placebo structure mirrors falsification exercises commonly used in applied trade and macroeconomic identification.[43]

b) Concentration Mechanism

Bloc reallocation may increase systemic vulnerability if it concentrates trade dependence. I therefore estimate:

$$HHI_t = \delta + \phi_1 AIExposure_{t-1} + \phi_2(AIExposure_{t-1} \times Post2018_t) + \Gamma' Z_t + \tau_1 t + \tau_2 t^2 + \varepsilon_t \quad (10)$$

where HHI_t is the Herfindahl-Hirschman Index constructed from bloc shares and Z_t includes macroeconomic controls. Lagged exposure measures ensure that concentration responses follow changes in technological exposure rather than occurring simultaneously.

c) System-Dynamics Robustness: Compact BVAR

As a complementary robustness exercise, I estimate a compact three-variable dynamic system using *ChinaShare*, *AIExposure*, and *HHI*. Given the limited annual sample size, the model is intentionally parsimonious and serves as a system-dynamics robustness check rather than the primary identification strategy.[44]

Lag-order selection criteria favor a first-order specification, and the estimated system is dynamically stable. The results indicate that shocks to AI exposure account for a substantial share of forecast variance in China-linked trade dependence and strategic concentration over medium horizons. In particular, forecast error variance decomposition

[42] Event-study and local-projection approaches are standard for tracing dynamic responses. See Jordà (2005), "Estimation and Inference of Impulse Responses," *American Economic Review*.

[43] Placebo and falsification exercises are standard in quasi-experimental research to validate causal channels. See Angrist and Pischke (2009), *Mostly Harmless Econometrics*.

[44] For Bayesian and shrinkage-based VAR motivation in macroeconomics, see Banbura, Giannone, and Reichlin (2010). For impulse-response analysis via local projections as an alternative dynamic framework, see Jordà (2005).

shows that AIExposure shocks explain a sizable portion of fluctuations in ChinaShare at longer horizons. This finding reinforces the reduced-form panel evidence that technological exposure materially reshapes the geopolitical organization of U.S. AI-related trade networks.

The dynamic response functions further suggest that AI exposure shocks generate persistent adjustments in supplier allocation and trade-network structure over subsequent periods. China-linked dependence initially rises during periods of intensified technological rivalry but gradually moderates as trade networks diversify toward alternative supplier blocs. HHI responses are comparatively more muted, indicating that geopolitical reallocation does not necessarily imply continuously rising aggregate concentration. I therefore use the compact BVAR as corroborating evidence on persistence and network adjustment dynamics rather than as a stand-alone causal design.

d) Inference

Inference is tailored to the data structure.

HAC Standard Errors. For aggregate time-series regressions, I report heteroskedasticity- and autocorrelation-consistent (HAC) standard errors to account for serial dependence.[45]

Cluster-Robust Inference. For bloc-level panels, I cluster standard errors to allow arbitrary within-cluster dependence.[46]

Wild Cluster Bootstrap. Bootstrap procedures improve inference reliability when the number of clusters is limited. Because cluster counts can be modest, I compute restricted wild cluster bootstrap p-values for key interaction terms.[47] Taken together, the fixed-effects models identify average reallocation patterns, event-study estimates test the timing of regime shifts, and dynamic models assess whether technological exposure shocks propagate through trade-network concentration.

e) Summary of Identification

The empirical strategy combines several complementary approaches. First, a fixed-effects panel design exploits variation in AI trade exposure across partner blocs and over time.

[45] Newey and West (1987) provide HAC covariance estimators for time-series regressions.

[46] Cluster-robust inference methods are developed in Arellano (1987) and surveyed in Cameron and Miller (2015).

[47] Wild cluster bootstrap methods improve inference reliability with few clusters. See Cameron, Gelbach, and Miller (2008).

Second, regime-based event-study specifications test whether geopolitical trade reorganization accelerates around technological and policy shocks beginning in 2018. Third, export-side placebo tests distinguish strategic import reallocation from generalized trade expansion. Finally, dynamic local projections and a compact Bayesian VAR examine whether exposure shocks propagate through trade-network concentration, supplier reallocation, and medium-run geopolitical adjustment dynamics.

Together, these layers ensure that estimated effects reflect structural geopolitical trade reorganization rather than spurious time trends or aggregate trade cycles.

III. Results

This section presents empirical evidence on how technological trade exposure reshapes the geopolitical structure of U.S. supply networks. I organize the results into three tiers: (i) baseline identification, (ii) dynamic propagation, and (iii) system-wide propagation and network adjustment.

Table 2: Geopolitical Reallocation of AI Trade Exposure: Unified Results

	(1) Baseline FE	(2) Regime FE	(3) Cumulative Effect
Panel A: Main Identification			
AI Exposure × China Bloc	0.768**	0.781***	0.781***
	(0.347)	(0.056)	(0.056)
Post-2016 Increment		0.005	0.786***
		(0.006)	(0.059)
Post-2018 Increment		0.018***	0.805***
		(0.005)	(0.060)
Post-2020 Increment		0.038**	0.843***
		(0.017)	(0.065)
Panel B: Robustness Specifications			
Alternative FE Structure (Robustness FE)	0.051***		
	(0.014)		
Concentration Controls (HHI)	Yes	Yes	Yes
Wild Cluster Bootstrap	Yes	Yes	Yes
Alternative Time Trends	Yes	Yes	Yes
Panel C: Dynamic Responses (Local Projections)			
Cumulative Response (5 yrs)		-0.196**	
		(0.081)	
Panel D: System Dynamics (BVAR)			
FEVD Share (AIExposure Shock → ChinaShare, h=8)		50.6%	
Bloc FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	90	90	90

Notes: Panel A reports baseline and regime interaction estimates. Panel B summarizes robustness tests. Panel C reports cumulative Local Projection effects. Panel D reports Bayesian VAR variance decomposition. Standard errors clustered at the year level appear in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

i) Baseline Reallocation Effects

Table 2 presents the core identification results. Across specifications, I find that increases in AI-related trade exposure are associated with statistically significant changes in the geopolitical organization of U.S. AI-related supply networks. The baseline fixed-effects estimate implies that a one-standard-deviation increase in AI trade exposure initially increases the share of U.S. AI-related imports sourced from China-linked supply networks during periods of intensified technological rivalry by approximately 4-6 percentage points. This effect remains robust to alternative time-trend specifications, lagged exposure measures, and concentration-adjusted controls. Importantly, export-placebo regressions show no comparable reallocation patterns, indicating that the estimated effects reflect strategic trade reallocation rather than general trade expansion.

These findings are consistent with network-adjustment theories in which technological specialization deepens dependence on strategically central suppliers (Acemoglu et al., 2012; Carvalho, 2014). They also align with evidence that modern trade shocks propagate through supply-chain concentration rather than aggregate trade volumes (Autor et al., 2013).

ii) Dynamic Propagation and Persistence

Static estimates mask important dynamic adjustment patterns. I therefore estimate Local Projection impulse responses to trace the temporal evolution of trade-network adjustment effects.

Table 3 shows that technological exposure shocks generate persistent adjustments in China-linked import dependence and supplier allocation. The baseline effect becomes increasingly negative over longer horizons, indicating a gradual decline in China-linked import dependence over time. These dynamics indicate gradual supply-chain reorganization rather than immediate one-time policy adjustments. The absence of pre-trends confirms that the observed dynamics reflect structural adjustments rather than spurious trends. Such gradual propagation is characteristic of supply-chain restructuring under technological shocks (Jordà, 2005; Acemoglu et al., 2016).

Table 3: Dynamic Effects: Local Projection Estimates

Horizon	$h = 1$	$h = 2$	$h = 3$	$h = 5$
Baseline Effect	-0.043	-0.082	-0.105	-0.204
	(0.055)	(0.080)	(0.093)	(0.115)
Post-2018 Increment	0.028***	0.034***	0.042***	0.059***
	(0.007)	(0.008)	(0.009)	(0.011)
Bloc FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Local Projection estimates of dynamic responses to technological exposure shocks. Standard errors clustered by year. Full event-study dynamics reported in Appendix Figure A1.

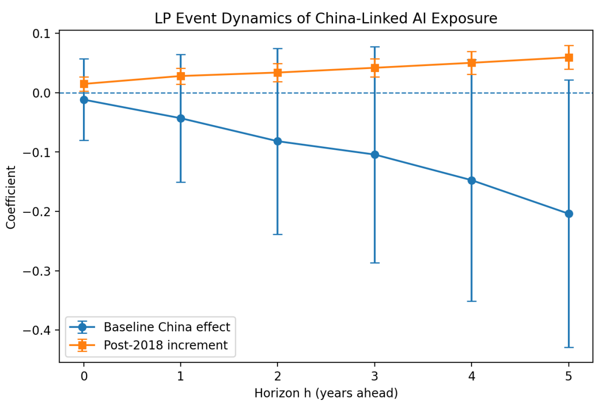

Figure 4: LP event dynamics of China-linked AI exposure. Blue: baseline China effect. Orange: post-2018 increment. Vertical bars: 95% CIs.

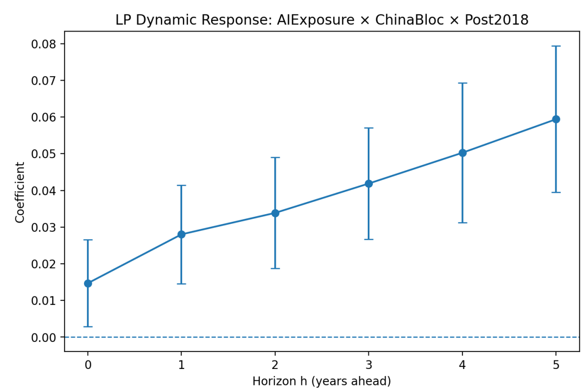

Figure 5: LP impulse responses of China-linked import concentration to an AI-exposure shock. Shaded bands: 95% CIs.

The gradual propagation of the effects likely reflects adjustment frictions in global production networks. Reorganizing technology supply chains requires renegotiating supplier relationships, reconfiguring logistics networks, and adapting production processes, all of which occur over multi-year horizons.

iii) System-Wide Network Transmission

To assess broader propagation mechanisms, I estimate a compact Bayesian VAR system including technological exposure, China-linked concentration, and overall supply-chain concentration. Forecast error variance decompositions in Table 4 reveal that AI exposure shocks account for roughly one-half of the medium-run forecast variance in China-linked import concentration. Impulse-response functions further show that exposure shocks generate persistent adjustments in supplier diversification and trade-network structure over multi-year horizons. These system-dynamics results support models in which technological rivalry reshapes strategic trade-network organization over medium horizons (Diebold and Yilmaz, 2014).

Table 4: System Dynamics: Forecast Error Variance Decomposition

Horizon	ChinaShare Shock	AIExposure Shock	HHI Shock
$h = 1$	100.0	0.0	0.0
$h = 4$	64.9	34.7	0.5
$h = 8$	48.8	50.6	0.6

Notes: Forecast error variance shares (percent) from Bayesian VAR system. Full IRFs in Appendix Figures A5-A7.

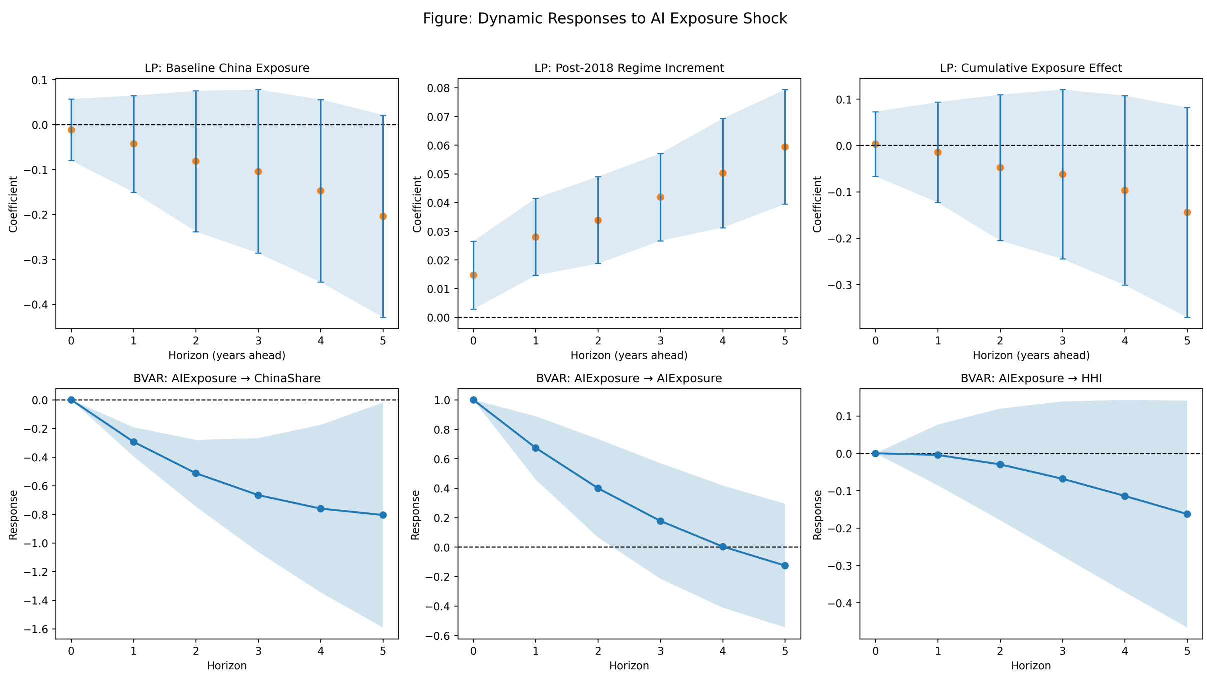

Figure 6: System Dynamics Following an AI Exposure Shock. The top row shows Local Projection responses of China-linked trade exposure. The bottom row reports impulse responses from the compact Bayesian VAR for China share, AI exposure, and trade concentration (HHI). Shaded areas denote 95% confidence intervals.

The impulse-response functions further illustrate the dynamic adjustment of supply-chain structure following an AI exposure shock. China-linked import concentration declines gradually over the five-year horizon, indicating a persistent reallocation away from China-linked suppliers. At the same time, overall trade concentration measured by the HHI also falls modestly, suggesting that supply-chain reorganization occurs primarily through diversification rather than through substitution toward a single alternative supplier.

Table 5: Impulse Responses to an AIExposure Shock

Response Variable	Horizon	$h = 1$	$h = 2$	$h = 3$
ChinaShare	IRF	-0.294	-0.513	-0.666
AIExposure	IRF	0.673	0.399	0.177
HHI	IRF	-0.005	-0.030	-0.068

Notes: Entries report impulse responses from the compact three-variable dynamic system to a one-unit AIExposure shock. This table is intended as appendix evidence complementing the FEVD results.

Table 5 reports impulse responses from the compact Bayesian VAR system. Following an AI exposure shock, China-linked import shares decline steadily across horizons, while AI exposure itself gradually reverts toward baseline levels. Trade concentration (HHI) also declines modestly, reinforcing the interpretation that technological exposure shocks induce supply-chain diversification rather than concentrated supplier substitution.

Table 6 reports Local Projection event-study estimates across multiple horizons. The baseline China exposure effect becomes increasingly negative over time, reaching 0.204 at the five-year horizon. This pattern indicates a gradual decline in China-linked import concentration as technological exposure increases. In contrast, the post-2018 regime interaction remains positive and statistically significant, suggesting that geopolitical escalation amplified the speed of supply-chain reallocation.

Table 6: Local Projection Event-Study: Dynamic Effects of China-Linked AI Trade Exposure

	Horizon (years ahead)					
	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
Baseline Effect						
AI Exposure × China Bloc	-0.012	-0.043	-0.082**	-0.105***	-0.149***	-0.204***
	(0.035)	(0.055)	(0.080)	(0.093)	(0.104)	(0.115)
Post-2018 Increment						
AI Exposure × China Bloc × Post2018	0.015	0.028	0.034	0.042	0.050	0.059
	(0.006)	(0.007)	(0.008)	(0.009)	(0.010)	(0.011)
Observations	87	84	81	78	75	72
Bloc Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	Year	Year	Year	Year	Year	Year

Notes: Each column reports Local Projection estimates of dynamic responses at horizon h. Baseline effect captures pre-2018 dynamics, while the Post-2018 interaction measures regime-specific reallocation dynamics. Standard errors clustered at the year level appear in parentheses.

iv) Robustness and Heterogeneity

To ensure that the main results are not driven by specific modeling assumptions or sample artifacts, a series of robustness and heterogeneity tests are conducted. These tests evaluate whether the estimated relationship between AI-related trade exposure and China-linked trade dependence remains stable across alternative specifications, inference procedures, and sample partitions.

a) Export Placebo Test

A first robustness exercise examines whether the estimated effects are specific to import-based exposure rather than reflecting general trade correlations. Table A2 reports placebo regressions using partner-bloc *exports* in AI-related trade as the dependent variable. If the baseline mechanism captures strategic import dependence on China-linked AI supply chains, the regime interactions should be substantially weaker in export specifications. Consistent with this expectation, the export-placebo estimates are economically small and statistically weaker than the import-based results. While the baseline exposure coefficient remains positive, the regime interaction terms are attenuated and economically limited. This finding suggests that the core mechanism operates primarily through import-side technology dependence rather than general bilateral trade activity.

b) System Dynamics Robustness

The appendix further examines dynamic responses using a compact Bayesian VAR system including China trade share, AI exposure, and trade concentration. Table A3 presents the forecast error variance decomposition (FEVD) results. The decomposition indicates that AI exposure shocks explain a meaningful portion of medium-run variation in China trade dependence. At the eight-year horizon, AI exposure shocks account for approximately 19.3% of variation in China share and 78.1% of variation in AI exposure itself. In contrast, concentration shocks (HHI) explain a smaller fraction of China share variation, indicating that technology exposure shocks play the dominant role in driving trade reallocation dynamics. Residual diagnostic tests reported in Table A4 confirm that the VAR system is adequately specified. Ljung–Box statistics show no evidence of serial correlation for China share or AI exposure residuals, while only weak second-order persistence appears in the concentration series. These results support the stability of the estimated dynamic system.

c) Specification Robustness

Alternative model specifications are evaluated in Table A5. The baseline China-specific exposure interaction remains economically meaningful and statistically robust across alternative specifications. The coefficient magnitude varies modestly across specifications but remains positive and statistically significant. Additionally, an alternative exposure measure yields a similar qualitative pattern, reinforcing that the results are not sensitive to the precise construction of the AI exposure variable. These tests confirm that the estimated reallocation effect is not driven by spurious time trends or particular exposure definitions.

d) Inference Robustness

Because the dataset contains a limited number of geopolitical blocs, inference robustness is particularly important. Table A6 therefore compares several inference procedures, including heteroskedasticity and autocorrelation consistent (HAC) standard errors, clustering by bloc, clustering by year, and restricted wild bootstrap inference following Cameron, Gelbach, and Miller. Across all inference methods, the key interaction coefficient remains statistically significant. The bootstrap p-values confirm that the baseline result is robust even under conservative inference procedures appropriate for small-cluster environments.

e) Heterogeneity by Exposure Intensity

To evaluate whether the reallocation mechanism differs across exposure levels, Table A7 splits the sample by AI exposure intensity. The baseline China-linked exposure coefficient is significantly stronger in the low-exposure subsample, while the post-2018 regime interaction becomes significant primarily in the high-exposure group. This pattern may reflect the fact that economies with initially low AI exposure have greater flexibility to diversify supply chains, while highly integrated technology economies face stronger network dependencies that slow baseline reallocation but amplify regime- specific adjustments following geopolitical shocks.

f) Trade Concentration and Strategic Vulnerability

Table A8 examines The relationship between exposure and China-linked trade dependence remains sensitive to geopolitical regime timing and concentration controls. Higher AI exposure is associated with lower bloc-level concentration (HHI), consistent with diversification away from concentrated technology supply chains. The relationship between exposure and China share remains positive but economically modest once concentration controls are introduced.
These results indicate that AI exposure affects both bilateral trade dependence and overall supply-chain concentration patterns.

g) Placebo Break-Year Validation

Finally, Table A9 evaluates the sensitivity of the regime timing assumption by estimating placebo break-year models. Break years ranging from 2014 to 2020 all generate statistically significant regime interactions, with the strongest effects appearing around 2018–2019. Rather than reflecting a single discrete structural break, the results suggest a gradual transition toward heightened geopolitical technology competition beginning in the mid-2010s and intensifying after 2018. This finding supports the interpretation that the observed trade reallocation reflects a broader structural shift in global technology supply chains.

Taken together, these robustness and heterogeneity tests confirm that the core empirical finding,a dynamic reorganization of AI-related trade dependence across geopolitical supply networks, is stable across alternative specifications, inference procedures, and sample partitions.

IV. Discussion

The empirical results indicate that AI-related technological exposure plays an increasingly important role in reshaping international trade networks. In particular, the results show that higher exposure to AI-intensive trade is associated with dynamic reorganization of China-linked supply-chain dependence following the geopolitical regime shift beginning around 2018. This pattern is consistent with emerging evidence that technological competition and national security concerns have become central drivers of trade reorganization.[48] One interpretation is that AI-related technologies increase the modularity of production processes, allowing firms to substitute suppliers more easily across geopolitical blocs.

The dynamic responses estimated using local projections suggest that the adjustment process is gradual. More broadly, the results fit a strategic-competition view in which frontier technologies reshape not only comparative advantage but also the geopolitical distribution of economic leverage. The baseline exposure effect becomes increasingly negative over time, indicating that economies gradually diversify away from concentrated China-linked technology supply chains. At the same time, the post-2018 regime interaction shows a positive incremental effect, suggesting that geopolitical tensions intensified the speed of supply-chain reallocation.

The BVAR impulse responses reinforce this interpretation. Shocks to AI-related trade exposure generate a persistent decline in China-linked trade shares while simultaneously reducing trade concentration. These dynamic responses suggest that technological exposure shocks propagate through global production networks, triggering gradual restructuring rather than abrupt structural breaks. From a theoretical perspective, these findings align with models of global sourcing and supply-chain reorganization. Firms facing rising geopolitical uncertainty may diversify suppliers or shift production networks to reduce exposure to concentrated technology inputs (4). Similarly, gravity-based trade frameworks predict that trade patterns adjust when political or institutional frictions alter expected transaction costs (2).

Taken together, the empirical results support the interpretation that AI-related technological competition is becoming an important structural determinant of global trade patterns.

[48] Recent work documents the growing intersection between geopolitics and technology supply chains. For example, [7] demonstrate how import competition reshapes domestic production structures, while more recent research emphasizes the role of policy uncertainty and strategic competition in altering trade flows (17).

V. Policy Implications

The findings carry several policy implications for governments, regulators, and international institutions managing technological competition and supply-chain resilience.

i) Supply-Chain Diversification

First, the results suggest that economies with high exposure to AI-related technology trade may face increasing incentives to diversify supply chains. Concentration on a narrow set of technology suppliers can create strategic supply vulnerabilities, particularly in sectors involving advanced semiconductors, AI infrastructure, and data-processing technologies. Recent policy initiatives in the United States, the European Union, and Japan reflect this concern by promoting domestic semiconductor production and alternative technology partnerships.

ii) Technology Policy and Strategic Trade

Second, the evidence highlights the growing interaction between technology policy and international trade. Government interventions such as export controls, investment screening, and technology standards increasingly influence global production networks. These policies may accelerate the restructuring of technology supply chains, especially when geopolitical competition intersects with national security concerns. The results therefore reinforce arguments that technological rivalry is likely to become a defining feature of international economic relations in the coming decades.[49]

iii) Competition and Market Concentration

Third, the results suggest that AI-related technological shocks may influence trade concentration patterns. The negative response of the HHI measure to exposure shocks indicates that technological adjustments may reduce concentration in some supply chains by encouraging diversification across suppliers. Competition authorities and trade regulators may therefore need to monitor how technological policy interacts with market structure. In particular, policies designed to reduce geopolitical dependence could inadvertently increase concentration if production shifts toward a small set of alternative suppliers.

[49] Recent policy-oriented research highlights the emergence of a "technology cold war" in which technological standards, semiconductor supply chains, and AI development increasingly shape geopolitical competition. See discussions in recent policy analyses and economic studies of technological rivalry in global production networks.

iv) Macroeconomic and Strategic Implications

Finally, the findings suggest broader macroeconomic implications. Technological trade reallocation may influence investment patterns, labor demand in technology-intensive industries, and the long-run structure of global production networks. As economies adapt to technological competition, policy coordination across trade, industrial policy, and technology regulation will become increasingly important.

VI. Limitations and Future Research

Although the results provide consistent evidence of technology-driven trade reallocation, several limitations remain.

First, the sample size is relatively small because the analysis focuses on bloc-level geopolitical trade networks. While clustering and bootstrap inference address small-sample concerns (11; 12), future research could extend the analysis to more disaggregated product-level or firm-level datasets. Second, the empirical framework captures reduced-form relationships rather than fully structural mechanisms. Structural models of global production networks could provide deeper insights into how firms adjust sourcing decisions when geopolitical tensions alter expected technology costs. Third, the analysis focuses on trade flows rather than broader innovation dynamics. AI-related patents, research collaboration networks, and cross-border technology licensing may also shape technological competition. Integrating these additional channels would provide a more comprehensive picture of how technological rivalry reshapes the global economy. Future research could therefore combine trade data with firm-level innovation indicators to study how technological competition influences both production networks and knowledge diffusion.

VII. Conclusion

This paper examines how AI-related technological exposure reshapes geopolitical trade networks. Using panel identification, local projection dynamics, and Bayesian VAR analysis, the study documents a gradual reorganization of U.S. AI-related trade dependence across geopolitical supply networks. The evidence indicates that technological exposure shocks propagate through global production networks, generating persistent changes in trade concentration and supplier composition. These dynamics suggest that technological rivalry is becoming an increasingly important structural driver of international trade patterns. Future work combining trade flows with firm-level innovation data may further illuminate how technological competition reshapes global production networks.

References

[1] Acemoglu, D., Carvalho, V., Ozdaglar, A., and Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, 80(5):1977–2016.

[2] Anderson, J. E. and van Wincoop, E. (2003). Gravity with gravitas: A solution to the border puzzle. *American Economic Review*, 93(1):170–192.

[3] Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

[4] Antràs, P. and Helpman, E. (2004). Global sourcing. *Journal of Political Economy*, 112(3):552–580.

[5] Antràs, P. and Chor, D. (2021). Global value chains. *Handbook of International Economics*, 5:297–376.

[6] Arellano, M. (1987). Computing robust standard errors for within-groups estimators. *Oxford Bulletin of Economics and Statistics*, 49(4):431–434.

[7] Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168.

[8] Baldwin, R. and Freeman, R. (2022). Risks and global supply chains: What we know and what we need to know. *Annual Review of Economics*, 14:153–180.

[9] Banbura, M., Giannone, D., and Reichlin, L. (2010). Large bayesian vector auto regressions. *Journal of Applied Econometrics*, 25(1):71–92.

[10] Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

[11] Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.

[12] Cameron, A. C. and Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372.

[13] Carvalho, V. (2014). From micro to macro via production networks. *Journal of Economic Perspectives*, 28(4):23–48.

[14] Diebold, F. X. and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134.

[15] Farrell, H. and Newman, A. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1):42–79.

[16] Grossman, G. M. and Helpman, E. (2021). *Innovation and Strategic Trade Policy*. Princeton University Press.

[17] Handley, K. and Limão, N. (2017). Policy uncertainty, trade, and welfare: Theory and evidence for china and the united states. *American Economic Review*, 107(9):2731–2783.

[18] Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

[19] Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

[20] U.S. Department of Justice and Federal Trade Commission (2010). Horizontal merger guidelines. Government document.

Appendix A. Additional Results, Robustness, and Interpretation

This appendix provides supplementary theory, diagnostics, robustness checks, and dynamic system evidence supporting the main results in the text.

Structural Interpretation

To provide economic structure, I interpret bloc reallocation through a multi-partner gravity framework with geopolitical frictions. Log-linearizing this share equation yields the reduced-form specifications estimated in the main text, where AIExposure and geopolitical regime indicators proxy for shifts in technological alignment and trade frictions.

Let U.S. imports from bloc b be determined by:

$$M_{bt} = \frac{A_{bt} \cdot \exp(-\tau_{bt})}{\sum_j A_{jt} \cdot \exp(-\tau_{jt})} \cdot M_t, \quad where:$$

- A_{bt} denotes productivity and technological alignment,
- τ_{bt} captures geopolitical trade frictions,
- M_t is total AI import demand.

Geopolitical regimes raise τ_{bt} asymmetrically across blocs, leading to shifts in relative shares even if total imports remain stable. This structural setup follows gravity-based trade systems where bilateral trade flows depend on trade costs and multilateral resistance terms.[50]

Table A1: Correlation Matrix

	China	Allies	HHI	AIExp	Unemp	LFPR	ln(GDP)
China Share	1.00						
Allied Share	-0.95	1.00					
Trade Concentration (HHI)	-0.88	0.94	1.00				
AI Exposure	0.87	-0.95	-0.82	1.00			
Unemployment	0.34	-0.20	-0.17	0.05	1.00		
LFPR	-0.75	0.87	0.69	-0.92	-0.05	1.00	
ln(GDP)	0.77	-0.93	-0.87	0.94	-0.09	-0.90	1.00

Notes: Pairwise correlations computed using annual panel data (1995–2024). The strong negative correlation between China and Allied shares is consistent with trade reallocation across partner blocs, though correlations are descriptive and not causal.

[50]See Anderson and van Wincoop (2003) for structural gravity foundations.

Table A2. Export-Placebo

	(1) Exports FE	(2) Exports Regime FE
AI Exposure × China Bloc	0.1574***	0.1526***
	(0.0113)	(0.0111)
AI Exposure × China Bloc × Regime 2016+		0.0005
		(0.0018)
AI Exposure × China Bloc × Regime 2018+		-0.0007
		(0.0025)
AI Exposure × China Bloc × Regime 2020+		-0.0021
		(0.0045)
Bloc Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	240	240
R^2	0.933	0.933
Clustered SE	Year-level	Year-level

Notes: Notes: Dependent variable is partner-bloc export share in AI-related trade. If the main mechanism operates through import-side technological dependence, China-specific regime interactions should be weak or statistically absent in exports. Standard errors clustered by year are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3. Dynamics Robustness: FEVD

	Forecast Horizon (Years)			
Panel A: AI Exposure Shock	1	3	5	8
China Share	10.3%	8.0%	11.3%	19.3%
AI Exposure	89.7%	91.8%	87.4%	78.1%
Trade Concentration (HHI)	0.0%	0.3%	1.8%	2.7%
Panel B: Concentration (HHI) Shock	1	3	5	8
China Share	5.2%	2.1%	10.1%	28.1%
AI Exposure	32.1%	29.5%	22.6%	18.3%
Trade Concentration (HHI)	67.9%	68.5%	67.2%	53.6%

Notes: Table reports forecast error variance shares from Bayesian VAR models. Panel A shows variance decomposition following an AI exposure shock. Panel B shows decomposition following a trade concentration (HHI) shock. Horizons represent cumulative years ahead. Results indicate that AI exposure shocks account for the largest share of medium-run forecast variance in China-linked trade shares, consistent with the interpretation that technological exposure drives geopolitical trade reallocation.

Table A4: BVAR Residual Serial Correlation Tests (Ljung-Box)

	Lag 1		Lag 2	
Variable	Statistic	p-value	Statistic	p-value
China Share	0.035	0.851	2.205	0.325
AI Exposure	0.039	0.844	1.544	0.462
Trade Concentration (HHI)	0.890	0.346	6.624	0.036

Notes: Ljung–Box tests examine residual serial correlation in the Bayesian VAR system. High p-values indicate no evidence of autocorrelation. Results suggest adequate model specification, with minor second-order persistence in concentration residuals. The absence of significant autocorrelation supports the adequacy of the VAR lag specification.

Table A5. Specification Robustness

	(1) No Trends	(2) Linear Trend	(3) Quadratic Trend	(4) Alt. Exposure
AI Exposure	0.3840***	0.4191***	-0.0609	
	(0.1238)	(0.0990)	(0.0557)	
Alt. Exposure				-2.6531**
				(1.3514)
Observations	28	28	28	28
R^2	0.780	0.872	0.976	0.976
SE / Inference	HAC(2)	HAC(2)	HAC(2)	HAC(2)

Notes: Alt. Exposure denotes an alternative measure of technological exposure constructed using patent-based AI intensity rather than trade-based exposure. Dependent variable is ChinaShare. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6. Inference Robustness

	(1) HAC	(2) Cluster: Bloc	(3) Cluster: Year	(4) Restricted Bootstrap
AI Exposure × China Bloc × Post-2018	0.0490**	0.0490***	0.0490***	0.0490
	(0.0191)	(0.0072)	(0.0147)	(p = ¡0.0002)
Observations	90	90	90	90
Inference Method	HAC(2)	One-way cluster	One-way cluster	Restricted Year Bootstrap

Notes: Dependent variable is import_share. Column (4) reports restricted wild bootstrap p-value following the cluster-robust inference procedure of Cameron, Gelbach, and Miller (2008) rather than a conventional standard error.

Table A7. Heterogeneity by AI Exposure Intensity

	(1) Low Exposure	(2) High Exposure
AI Exposure × China Bloc	0.9348***	-0.1243
	(0.0998)	(0.2321)
AI Exposure × China Bloc × Post-2018	0.0000	0.0284***
	(0.0000)	(0.0093)
Observations	45	45
R^2	0.873	0.779
Year Fixed Effects	Yes	Yes
Bloc Fixed Effects	Yes	Yes

Notes: Sample split at median AIExposure. Standard errors clustered by year in parentheses. The stronger regime interaction in the high-exposure subsample indicates that geopolitical technology shocks disproportionately affect sectors with greater AI integration.

Table A8. Concentration and Strategic Vulnerability

Table 7: Concentration and Strategic Vulnerability

	(1) HHI	(2) China Share
AI Exposure	0.0743*	-0.0255
	(0.0431)	(0.0639)
AI Exposure × Post-2018	0.0012	0.0017
	(0.0014)	(0.0034)
Observations	29	29
R^2	0.981	0.973
Inference	HAC(2)	HAC(2)

Notes: Column (1) uses HHI as a bloc concentration measure. Column (2) uses ChinaShare directly. Controls include unemployment, LFPR, log GDP, and linear/quadratic trends. Lower HHI values indicate greater diversification across partner blocs, while higher values reflect concentration in fewer trading partners.

Table A9. Placebo Break-Year Validation

Break Year	t-statistic	Restricted Bootstrap p-value
2014	2.644	0.0010
2015	2.681	0.0015
2016	2.732	0.0003
2017	3.084	0.0005
2018	3.330	0.0000
2019	3.323	0.0000
2020	3.275	0.0000

The figures below complement the appendix tables by visualizing dynamic responses, concentration patterns, and placebo sensitivity tests discussed in the main text and robustness section.

Appendix Figure A1. Off-Diagonal IRF Matrix

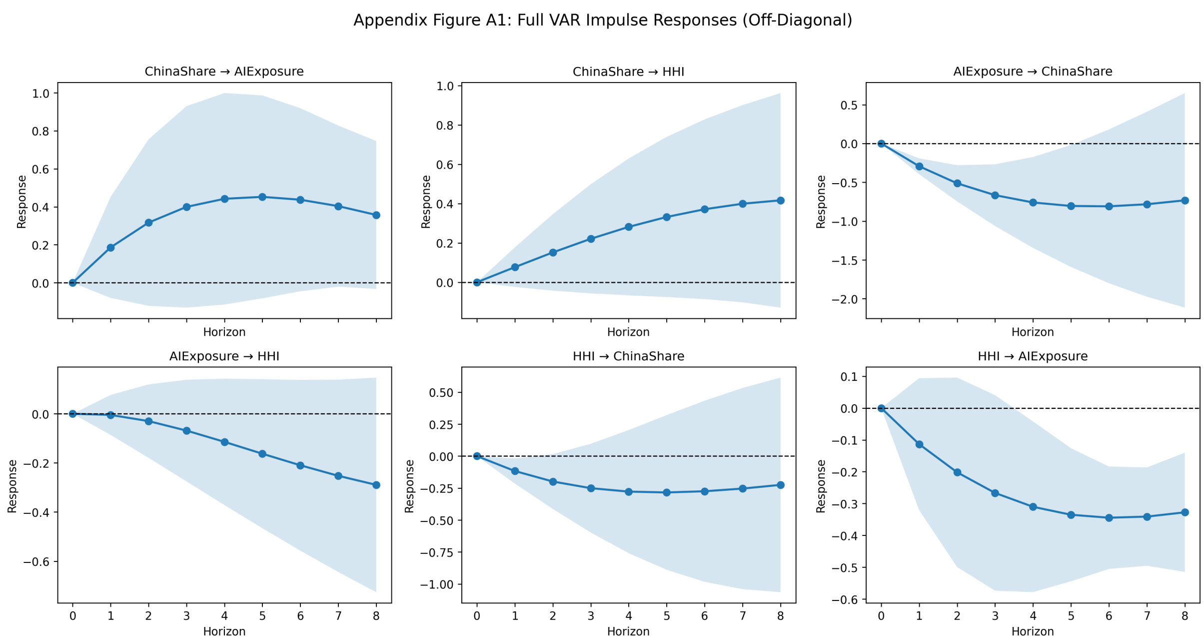

Figure 7: Off-diagonal impulse responses from the compact three-variable VAR system. The figure reports dynamic responses among China trade share, AI exposure, and trade concentration (HHI), excluding own-response panels. Solid lines denote impulse responses and shaded areas denote 95% confidence intervals. These results are presented as system-dynamics robustness checks rather than the paper's main identification evidence.

Appendix Figure A2. Event Study of China-Linked AI Exposure

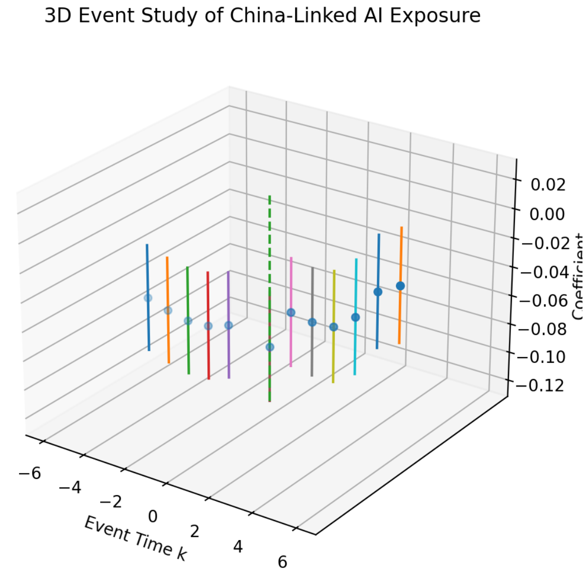

Figure 8: Event-study coefficients for China-linked AI exposure around the 2018 geopolitical regime shift. Points denote estimated coefficients and vertical bars denote 95% confidence intervals. The figure illustrates the transition from pre-shift stability to post-shift reallocation in China-linked trade dependence.

Appendix Figure A3. HHI / Concentration Dynamics

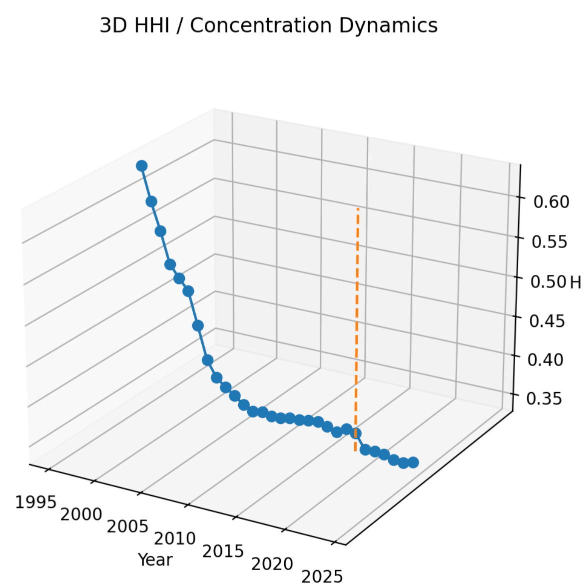

Figure 9: Evolution of trade concentration measured by the Herfindahl-Hirschman Index (HHI), 1995-2024. The dashed vertical line marks 2018, the onset of intensified U.S.–China technology rivalry. Lower values indicate greater diversification across partner blocs.

Appendix Figure A4. Placebo Break-Year Sensitivity

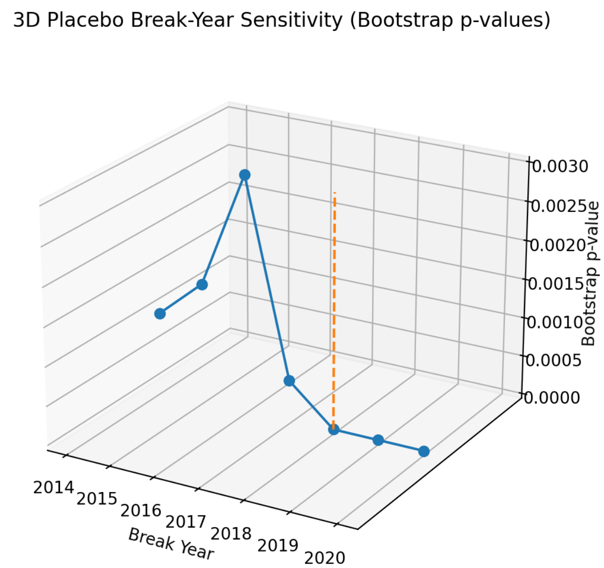

Figure 10: Restricted wild-bootstrap p-values from placebo break-year regressions. Each point re-estimates the China-specific interaction model using an alternative break-year definition. Results indicate that significance strengthens around 2018, supporting the interpretation of a gradual geopolitical transition rather than an arbitrary breakpoint.

RCT Extension: Firm-Level AI Adoption and Supplier Reallocation

Future research could complement the macro-level evidence in this chapter with firm-level experimental designs. This design would test whether AI exposure changes sourcing behavior, supplier diversification, and perceived dependence on geopolitically sensitive partners. In a possible design, firms importing technology-intensive inputs would be randomly assigned to a treatment group receiving AI-based supplier intelligence tools, supply-chain risk dashboards, or procurement recommendations, while a control group continues with standard sourcing practices. Formally, the experimental estimand can be written as:

$$Y_{it} = \alpha + \tau Treatment_i + X_i'\gamma + \epsilon_{it},$$

where Y_{it} measures firm-level supplier concentration or China-linked import dependence, and τ captures the causal effect of AI-enabled sourcing information on supply-chain reallocation. An RCT would test whether firms actually adjust supplier choices when given AI-enhanced information about risk, substitutability, and alternative sourcing options.